

Kugire Data Curation for the Kokinwakashu TEI/XML Data

1. Introduction

Kugire — phrase breaks within waka poems — are a basic structural element of waka, yet no standardized, publicly available kugire dataset exists. For TEI/XML encoding of waka, each project must annotate from scratch.

[atuko315/program-to-divide-31-syllable-Japanese-poem](https://github.com/atuko315/program-to-divide-31-syllable-Japanese-poem)¹ is one of the few existing kugire resources: a program to identify kugire with a dataset for the Hyakuninissu, later applied to the Kokinwakashu and compared against Asaoka (2008) and Kami (1985). The dataset, however, cannot capture the interpretive nature of kugire (see Kami, 1985), which admits no single correct answer. Asaoka (2007) proposes two criteria — semantic independence of core sentences (necessary) and a clear grammatical break (sufficient) — both of which remain open to interpretation. A kugire dataset should therefore collect annotations from multiple sources, each with documented criteria.

This project adds kugire as an interpretive annotation layer to the TEI/XML data of the Kokinwakashu, using rule-based and translation-based methods. Keeping kugire as a separate layer preserves the annotation criteria and any competing interpretations. The data supports analysis of waka structure, enjambment, diachronic change of kugire position, and its relation to poetic style.

2. Data sources

2.1. Base data

The base data is a lexically extended version of the TEI data of the Karoku second-year manuscript of the Kokinwakashu (Ikuura et al., 2023). The extended dataset maps tokenization from the Hachidaishu Vocabulary Dataset (Hodošek and Yamamoto, 2022) onto this segmented text, forming the basis for the kugire annotation.

2.2. Reference data

For the rule-based method, we use the Hachidaishu Part-of-Speech Dataset (Yamamoto et al., 2024a), which provides conjugation and part-of-speech data for each token. For the translation-based method, we use the explanatory translation of the Kokinwakashu by (Kaneko, 1933), the earliest twentieth-century explanatory translation of the anthology, digitized and published on Zenodo (Yamamoto et al., 2024b).

¹<https://github.com/atuko315/program-to-divide-31-syllable-Japanese-poem>.

3. Output Data

The output is `kokin-kugire.xml`, a TEI/XML file derived from the base data by appending kugire annotations to each poem. The original text, segmentation, and all existing TEI markup are preserved without modification; the annotation layer is additive.

Each kugire position is encoded as a `<k>` element placed after the final `<seg>` child of the corresponding line element `<l>`:

```
<l n="1" xml:id="n1">
  <seg>年の内に</seg>
  <seg>春はきにけり</seg>
  <seg>ひとゝせを</seg>
  <seg>こそとやいはむ</seg>
  <seg>ことしとやいはむ</seg>
  <k n="2" source="morph" cert="high"/>
  <k n="3" source="morph" cert="low"/>
  <k n="4" source="morph" cert="high"/>
  <k n="2" source="kaneko"/>
</l>
```

Figure 1: TEI/XML encoding of kugire annotations for poem 1 of the *Kokinwakashu*. Each `<k>` element marks a candidate break position. Morphological candidates carry a three-level certainty grade (`cert`: high, mid, low); translation-based candidates (`source`="kaneko") carry none.

The rule-based method identifies three candidate positions. The strongest (`cert`="high") is after the second segment, where *keri* appears in the conclusive form, ending the first complete sentence. A weaker candidate (`cert`="low") is also placed after the third segment. The LLM-assisted translation-based method, drawing on explanatory translation (Kaneko, 1933), identifies only the second segment as a kugire. The competing candidates are preserved independently.

The `@n` attribute is the 1-based segment index; `@source` is the evidence code; `@cert` (high / mid / low) grades morphological certainty for rule-based annotations and is omitted for translation-based ones.

Multiple `<k>` elements at the same position represent competing interpretations — from the same source at different grammatical strengths, or from different sources — and are preserved as-is. No normalization or deduplication is applied.

The `<k>` element is a project-internal tag pending a decision on the target TEI encoding.

4. Annotation pipeline

The annotation follows an LLM-assisted pipeline:

1. Rule-based suggestion of candidate kugire positions based on Yamamoto et al. (2024a)
2. LLM-assisted translation-based suggestion of candidate kugire positions based on Kaneko (1933 Yamamoto et al., 2024b)

3. Review and correction in a pop-up editor showing the original poem and its reference data
4. Rendering the confirmed annotation into the TEI data

The pipeline accelerates the annotation process. The per-poem review cycle reduces cognitive load and improves accuracy.

4.1. Rule-based (grammar-wise) kugire suggestion

Our method extends the program², which flags conclusive forms, copulas, and sentence-final particles as kugire, by outputting graded candidates — strong, moderate, and weak — rather than a single result.

Strong candidates are phrases ending in the conclusive form. For example, in the first poem of the Kokinwakashu, the past-tense auxiliary verb *keri* appears in the conclusive form, indicating a grammatical break after it (Figure 2-(1)).

Moderate candidates are phrases where the copula or information structure suggests a break, though not conclusively. In the second poem of the Kokinwakashu, the segments before and after *wo* are syntactically connected, yet the former presents a scene while the latter offers a conjectural statement about it — a shift that may indicate a break (Figure 2-(2)).

Weak candidates are phrases where a sentence-final particle suggests a break, but the surrounding segments remain semantically interdependent. In the third poem of the Kokinwakashu, the question particle in the second segment may suggest a break, yet the following segment answers that question — semantic interdependence that makes the break uncertain (Figure 2-(3)).

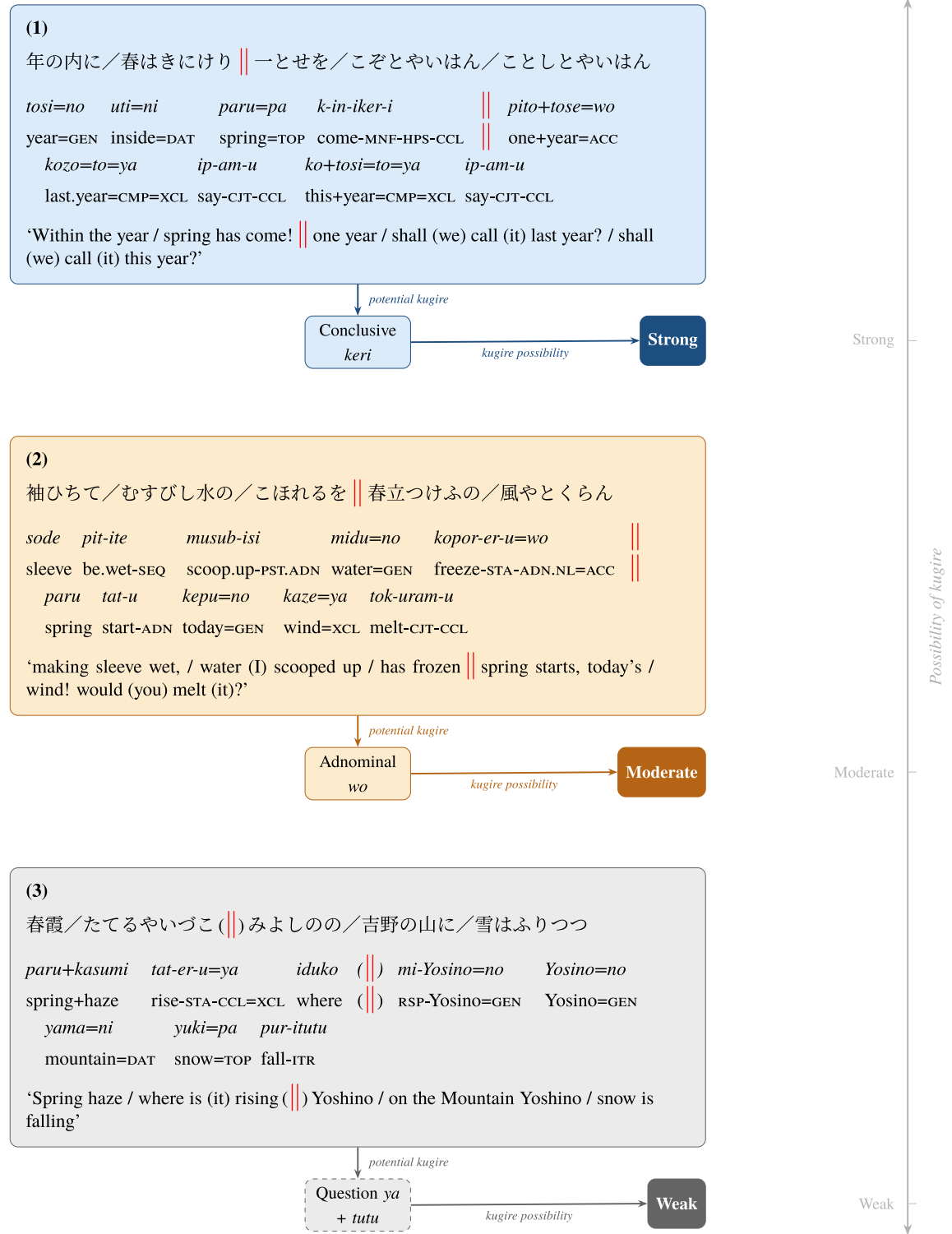
4.2. LLM-assisted translation-based kugire suggestion

We also use a large language model (LLM) to suggest candidate kugire positions from the explanatory translation of (Kaneko, 1933) as encoded by (Yamamoto et al., 2024b). Where the rule-based approach draws on morphological and grammatical features, the LLM-assisted translation-based approach draws on the contemporary Japanese translation, capturing breaks rooted in the translator's reading rather than grammatical form alone.

The LLM is qwen2.5, served locally via Ollama. The model receives the five segments of the original poem and the corresponding explanatory translation, and outputs a JSON object with three fields: `alignment` (step 1), `breaks` (step 2), and `positions` (the resulting candidate indices).

The prompt instructs the model to align segments to the translation, detect sentence boundaries in the translation, and map them back to segment positions — mirroring the workflow of a human annotator using a translation as reference. Three few-shot examples from real Kokinwakashu poems demonstrate this chain, covering no-kugire, second-segment, and fourth-segment cases. Valid positions from the JSON output are recorded independently of rule-based results, e.g., `<k n="2" source="kaneko"/>`.

²<https://github.com/atuko315/program-to-divide-31-syllable-Japanese-poem>.



Abbreviations: - = affix boundary; = = clitic boundary; + = compound boundary; ACC = accusative; ADN = adnominal; ADN = adnominal; CCL = conclusive; CJT = conjectural; CMP = complementizer; DAT = dative; GEN = genitive; HPS = hearsay past; ITR = iterative; MNF = manifestive; NL = nominalizer; PST = past; RSP = respect; SEQ = sequential; STA = stative; TOP = topic; XCL = exclamative

Figure 2: Three types of kugire annotation derived from morphological analysis: (1) strong, (2) moderate, and (3) weak candidates. Glossing follows Zisk (2023).

5. Conclusion

This project produced a TEI/XML dataset prototype of kugire annotations for the Kokinwakashu, derived from a lexically extended base and informed by both rule-based and LLM-assisted translation-based methods. The dataset preserves multiple interpretations of kugire, supporting analysis of waka structure and diachronic change.

6. References

- Asaoka Sumiaki** 浅岡純朗. (2008). Waka ni Okeru Bun no Kosei: Heian Waka no Tenkanki ni Kansuru Ichi Kosatsu/Sentence Structure in Waka: A Study on the Transitional Period of Heian Waka [和歌における文の構成—平安和歌の転換期に関する一考察—]. *Nishogakusha Daigaku Jinbun Ronso* / *Nishogakusha University Journal of Humanities*, **81**, pp. 130–40.
- Asaoka Sumiaki** 浅岡純朗. (2007). Waka no Kugire ni Kansuru Ichi Kosatsu: Atarashii Nintei Kijun Shian no Koso/A Study on Kugire in Waka: Toward a New Proposal for Recognition Criteria [和歌の句切れに関する一考察：新しい認定基準私案の構想]. *Nishogakusha Daigaku Jinbun Ronso* / *Nishogakusha University Journal of Humanities*, **78**, pp. 113–22.
- Hodošček, B. and Yamamoto, H.** (2022). Development of Datasets of the Hachidaishū and Tools for the Understanding of the Characteristics and Historical Evolution of Classical Japanese Poetic Vocabulary. In *Digital Humanities 2022 Conference Abstracts*. Tokyo, Japan.
- Ikuura Hiroyuki** 幾浦裕之., **Nagasaki Kiyonori** 永崎研宣. and **Kato Yumie** 加藤弓枝. (2023). Chokusen Wakashu no Kozo-ka to Teiji Shuho ni Kansuru Kokoromi: Karoku Ninenhon Kokinwakashu wo Jirei to shite/An Attempt at Structuring and Presenting Imperial Waka Anthologies: The Karoku Second-Year Manuscript of the Kokinwakashu as a Case Study [勅撰和歌集の構造化と提示手法に関する試み —嘉禄二年本『古今和歌集』を事例として—]. In *Proceedings of JINMONCOM 2023*. Information Processing Society of Japan, pp. 183–90.
- Kami Hiroyuki** 紙宏行. (1985). Shin-Kokin ni Okeru Sankugire no Hyogen Kozo/The Expressive Structure of Third-Line Breaks in the Shin-Kokinshū [新古今における三句切れの表現構造]. *Bunkyo Daigaku Joshi Tandai-bu Kenkyu Kiyo/Bunkyo University Women's College Bulletin*, **29**, pp. 10–23.
- Kaneko Motoomi** 金子元臣. (1933). *Kokinwakashu Hyoshaku: Showa Shimban/An Annotated Kokinwakashu: The New Showa Edition* [古今和歌集評釈：昭和新版]. Tokyo: Meijishoin.
- Yamamoto, H., Hodošček, B. and Chen, X.** (2024a). Hachidaishu Part-of-Speech Dataset. 10.5281/zenodo.13940187³.
- Yamamoto, H., Hodošček, B. and Chen, X.** (2024b). Kokinwakashu Hyoshaku by Motoomi Kaneko Translation Sentence Vocabulary Dataset. 10.5281/zenodo.13942707⁴.

³<https://doi.org/10.5281/zenodo.13940187>

⁴<https://doi.org/10.5281/zenodo.13942707>

Zisk, M. (2023). Glossing Glosses: Methods for Transcribing and Glossing Japanese Kundoku Texts. In Cinato, F.Lahaussois, A.and Whitman, J.B. (eds), *Glossing Practice, Comparative Perspectives*. Lexington Books, pp. 47–82.